

XTag

A narrative-intelligence platform.

It searches 21 open sources for a claim, hashtag or name, removes what isn't about it, works out who is pushing it and whether they are working together, and says what it does and does not know.

This document is everything a designer needs to redesign its interface: what it does, every field it produces, every state it can be in, and the rules its display must not break.

PRODUCT	XTag · xtag.up.railway.app
STACK	Flask + Unicorn on Railway · single-file vanilla-JS frontend
SOURCE OF FIGURES	Live run of <code>#covid1948</code> , 4 September 2026
REVISION	4 September 2026
CONTENTS	12 sections · data contract, scoring, states, constraints, open questions

What XTag is

You type a claim, a hashtag, a person or an organisation. XTag returns an assessment of the information environment around it, built from documents it collected and can show you.

It answers five questions, in this order:

1. **What is actually being said?** — distinct narratives and claims, not a feed of posts.
2. **Who is saying it?** — named actors, and the accounts carrying the volume.
3. **Are any of them working together?** — coordinated behaviour between accounts.
4. **Did it do anything?** — offline events, and whether reach was captured or manufactured.
5. **How much of this can I trust?** — what was collected, what failed, what is unknowable.

The fifth question is the differentiator and the hardest thing to design. Most tools in this space present a confident number. XTag's core commitment is that it will not present a number it cannot justify — the interface has to make an honest partial answer look competent rather than broken.

Who uses it

A counter-terrorism / OSINT analyst, working alone, who has to **defend a conclusion to someone else** — a government client, a supervisor, a report reader. That defence requirement drives everything: every figure needs to be traceable to documents, and every limitation needs to be visible before the analyst repeats the claim, not after.

Working session

Runs 5-20 queries in a sitting. Compares them. Comes back to one from last week and needs it exactly as it was.

Time pressure

Wants the shape of an answer in seconds and the full picture in under a minute. Currently ~8s to first screen, ~35s complete.

Output

The end product is usually a written assessment elsewhere. XTag needs to make evidence easy to lift, cite and quote.

The three principles the interface must serve

1 • Signal from noise

The user's own words for the product's purpose. On a live query, **52% of what was collected was not about the query** and was removed. The interface should make the filtering visible and reviewable — a user must be able to see what was dropped and why, because a filter you cannot audit is a filter you cannot trust.

2 • Intelligence, not raw information

A list of 96 posts is not intelligence. A judgement supported by 96 documents is. The screen's primary content should be a finding; the documents are its evidence, reachable in one click but not the first thing presented.

3 • Nothing shown twice

Repetition is a *property* of a claim — how many accounts carried it, across how many platforms — not a reason to render more rows. One wire story found by three collectors is one story. One claim posted by eleven accounts is one claim with eleven carriers. The backend already deduplicates and counts; the interface should present the count, not the copies.

How a search runs

Timings are measured from the live deployment, not estimated. The design implication is at the bottom of this section and it matters more than the numbers.

1	Collection	~8s	21 sources queried concurrently. Slowest source sets the time; stragglers are abandoned at a deadline and reported as failed rather than waited for.
2	Relevance gate	0.02s	Documents not about the query are removed. Rejected documents are kept for audit, with the reason each one went.
3	Deduplication	<0.1s	One article found by several collectors collapses into one record that absorbs the richest data from each.
–	First screen ships	~8s	Everything above is sent to the browser here. Documents, source mix, relevance report and counts are final and displayable. Everything below is still running.
4	Language + sentiment	~12s	Language detected, foreign-language previews translated, sentiment scored natively per language rather than after translation.
5	Analysis	~16s	Six concurrent model calls: narratives, entities, real-world events, velocity, coordination, propagation.
6	Assessment	<0.1s	Threat score, four risk dimensions, inauthenticity signals, audience and geography — all computed from stages 1–5.
7	Narrative tracking	~1s	This run's narratives are matched against previous runs of the same query, so a narrative keeps a stable identity over time.

THE DESIGN CONSEQUENCE

The screen is built in **two deliveries, ~25 seconds apart**. The first is complete and trustworthy on its own; the second adds interpretation. The interface must make the first delivery feel like a finished, usable state — not a loading skeleton — and must never render an unscored value as a zero. Sentiment that has not run yet is *not* "0 positive, 0 negative".

The 21 sources

Status is from the live deployment. A designer needs this because roughly half of any given run's sources return nothing, and that has to look normal rather than broken.

SOURCE	TYPE	STATUS	NOTES FOR DISPLAY
x	social	HEALTHY	Highest-yield source. Two passes (Top + Latest), deduplicated.
youtube	video	HEALTHY	Highest volume, lowest precision — 75% of its results are typically dropped by the gate.
google	web	HEALTHY	Web fallback; also the route to several social platforms via site-restricted search.
telegram	social	HEALTHY	A fixed list of 23 channels, not a search of Telegram. Coverage is deliberately narrow.
state_media	news	HEALTHY	Named adversary outlets by RSS. Documents are flagged <code>source_type: state_media</code> .
reddit	social	HEALTHY	
hackernews	social	HEALTHY	Fuzzy matching; most results are typically dropped.
mastodon	social	HEALTHY	
gnews	news	HEALTHY	
academic	academic	HEALTHY	Results are often unverified against the query — flagged as such.
gdelt	news	FAILING	The intended news backbone. Currently times out; a circuit breaker now skips it after repeated failures.
instagram	social	FAILING	
notebooklm	knowledge	FAILING	User's own knowledge base.
tiktok	social	INTERMITTENT	
bluesky	social	UNCONFIGURED	Needs credentials. Absence is configuration, not failure — worth distinguishing in the UI.
podcasts	audio	UNCONFIGURED	Needs credentials.
facebook · linkedin · pinterest · threads · tumblr	social	INDIRECT	Reached only via site-restricted web search, so yield is low and inconsistent.

On a representative run: **9 of 21 sources returned results, 3 failed, 9 returned nothing.** "Returned nothing" and "failed" are different facts and the interface currently conflates them.

The data contract

Everything the backend produces, by top-level key. This is what a design can draw on — nothing here needs to be built, and anything not here would need backend work.

PER DOCUMENT

FIELD	TYPE	WHAT IT IS
id	string	Stable hash of the URL.
platform	string	Which source found it.
source_type	enum	<code>social</code> · <code>news</code> · <code>state_media</code> · <code>academic</code>
title, excerpt	string	Original language.
title_en, excerpt_en	string	English translation, when the document is not English.
translated	bool	Whether what is shown is a translation. Must be visible to the reader.
language	string	Detected language code.
author, author_url	string	Handle for social, publishing domain for news.
url, thumbnail	string	
timestamp	ISO 8601	May be absent — a meaningful share of documents are undated.
credibility	enum	<code>high</code> · <code>state</code> · <code>low</code> · <code>unknown</code> , from a curated domain list.
engagement	int	Reactions + comments + shares. Deliberate acts only.
views	int	Impressions. Held separately and deliberately excluded from engagement.
engagement_breakdown	object	<code>{reactions, comments, shares, views}</code>
sentiment	float	−1 to +1. Also <code>s_claude</code> and <code>s_babel</code> — two engines, separately visible.
framing	enum	<code>fear</code> · <code>anger</code> · <code>hope</code> · <code>pride</code> · <code>grief</code> · <code>threat</code> · <code>disinformation</code> · <code>neutral</code>
relevance	float	0–1 score against the query.
relevance_basis	string	Plain-English reason it was kept, e.g. "exact match: 'covid1948'".
also_found_on	array	Other collectors that found the same document before deduplication.

RELEVANCE — THE FILTERING AUDIT

FIELD	WHAT IT IS
collected / kept / dropped	Counts before and after the gate.

FIELD	WHAT IT IS
noise_ratio	Proportion removed. Typically 0.5+.
canonical, surface_forms	The normalised query and every spelling matched. Useful to show what was actually searched for.
query_kind	hashtag · phrase · boolean
by_platform	Per source: collected, kept, dropped, verified, unverified.
group.filtered	Up to 20 rejected documents per source, each with its rejection reason — the audit trail.

NARRATIVES

Up to 8 clusters. Each: `label`, `count`, `framing`, `platforms[]`, `key_claim` (one sentence), `actors[]`, `velocity` (accelerating / stable / declining).

EVENTS – WHAT HAPPENED OFFLINE

Deliberately conservative; returns empty when the corpus does not evidence an event. Each: `label`, `date`, `location`, `kind` (`protest` · `violence` · `official_action` · `platform_action` · `investigation` · `other`), `evidence` (a quoted phrase from a real document), `confidence`.

This is the field that separates "a hashtag trended" from "a campaign did something", and it is currently under-used in the interface.

ENTITIES

`entities[]` — name, type, mentions, sentiment, and a supporting `quote` verified against the corpus. `edges[]` — from, to, relation. Also `insufficient_evidence`: a string explaining that the corpus was too thin to name actors, which is a different state from "no actors found".

COORDINATION

FIELD	WHAT IT IS
coordination_score	0-100 magnitude.
risk	low · medium · high — or unbanded . See rules.
baseline_ratio	Multiple of matched organic baseline, or <code>null</code> .
caveat	Present when unbanded. Explains why no band is shown.
clusters[]	Per cluster: <code>actors[]</code> , <code>actor_count</code> , <code>documents</code> , <code>cohesion</code> (0-1), <code>traits[]</code> , <code>examples[]</code> .

FIELD	WHAT IT IS
traits	Which behaviour links the accounts: <code>url · hashtag · text · time</code> . Shared links are the strongest evidence, shared wording the weakest.
trait_pairs	How many account pairs each behaviour found.
flagged_urls[]	Every document produced by clustered accounts. Not capped — the full set.
reach	<code>{manufactured, captured, manufactured_share, verdict, cluster_documents}</code> . Share may be <code>null</code> , meaning the corpus had no engagement figures — a data gap, not a zero.

NARRATIVE_TRACKING – IDENTITY OVER TIME

Narratives keep a **stable id across searches of the same query**. Each carries `label`, `members[]`, `first_seen`, `last_seen`, `observations`, `misses`, `peak_size`, and `history[]` (a time series of size).

`events[]` describes what happened between observations:

birth / death

First observed, or absent for three consecutive runs.

growth / contraction

Changed by more than $\pm 10\%$.

merge / split

Narratives converged, or one became several.

`drift` is the subtle one: the narrative kept its documents but its *subject* changed. Worth designing deliberately — it is how a tracked line silently stops being about what the analyst thinks it is about.

THREAT, RISK, AUDIENCE, INAUTHENTICITY

See the next section. Every score arrives with the weighted factors that produced it, each carrying `score`, `weight`, `contribution_pct`, `available`, `detail` and `reason`. **The current interface discards nearly all of this** — it renders a number and a bar. The explanation the user has repeatedly asked for already exists in the data.

DIAGNOSTICS ON EVERY RESPONSE

`timings` (per stage, in seconds) · `source_timings` (per source, with abandoned ones marked) ·
`degraded[]` (why the result is partial) · `engine_errors` (which analysis stages failed and why) ·
`deduplicated` (how many collapsed).

What each number means

The user's original complaint was that the scores "look good but I don't know what to do with them". Every number below has a defensible meaning; the interface has to carry it.

Threat score — 0 to 100

A weighted composite. The weights are fixed and worth surfacing, because they explain the number:

FACTOR	WEIGHT	MEASURES
volume	40	Coverage volume, log-scaled.
velocity	35	Whether it is still accelerating.
platforms	25	Spread across platforms.
state_media	20	Share carried by state-controlled outlets.
reach	12	Engagement volume.
tone	10	Hostile framing density.
credibility	8	Source credibility mix.
propagation	8	Cross-platform movement.

Bands: ≥ 75 critical · ≥ 55 high · ≥ 35 elevated · ≥ 18 moderate · below that, low. A band is suppressed when coverage is too thin to support one.

The four risk dimensions

Each answers a different question and reaches a different person. Each carries its own `score`, `band`, `coverage`, `factors[]` and a plain-English `rationale` string that the interface should simply print.

Reputational

Hostile framing (40) × geographic breadth (30) × acceleration (30). "Is this damaging, and is it spreading?"

Information integrity

Inauthentic amplification (45) × coordination (35) × state-media share (20). "Is this being manufactured?"

Operational

Whether this is disrupting something you run.

Physical security

A lexical flag that violent language is present and a human should look. **It does not predict violence and the card must say so.**

Confidence

How much to trust the assessment — *not* how bad the situation is. Driven by how many on-topic documents were found and how many signals were available. The distinction is not currently legible in the interface and is a recurring source of confusion.

Sentiment

Scored per document in its native language by two independent engines, then averaged. Above +0.25 positive, below −0.25 negative, between them neutral. **Unscored documents are reported separately from neutral ones** — a document nobody scored is not a document scored as neutral. Engine agreement percentage is available.

Coordination and reach

Coordination is measured between *accounts*, not documents, across four behaviours weighted by their published detection strength: shared links (0.72 AUC), shared tags (0.60), shared wording (0.52), shared timing (0.45). Reach is split into **manufactured** (engagement on documents by clustered accounts) and **captured** (everything else). Confirmed influence operations show 0.1–5.3% manufactured; a high manufactured share means the operation is talking to itself.

Screens today

What exists, and what each is for. GAP marks something the data supports but the interface does not yet do.

SCREEN	JOB	STATE
Search result	The main screen. Assessment above, evidence grid to the right.	NEEDS REDESIGN
Evidence grid	Every kept document as a card, filterable and sortable.	WORKS
Case history	Past searches, restorable exactly as captured, stored in the browser.	WORKS
Entity map	Actors and relationships as a canvas graph; hover reveals sources.	CRUDE
Timeline	Volume over time with brushing and spike annotation.	CRUDE
Watchlists	Saved queries re-run on a schedule, with alert rules and email reports.	WORKS, UNSURFACED
Printable dossier	A print-first report at <code>/report?q=...</code> , no JS.	WORKS, UNSURFACED
Entity dossier	One actor, everything known about it, across searches.	GAP
Coordination view	Clusters, behaviours, reach split as a screen of its own.	GAP
Narrative tracking	One line per narrative id over time. The data exists and is unused.	GAP
Claim view	Distinct claims with their carriers, rather than documents.	GAP
Comparison	Two queries, or one query at two times, side by side.	GAP

Non-negotiable display rules

These are enforced in the backend and tested. A design that breaks them undoes engineering work that exists specifically to prevent a class of error.

RULE 1 – NEVER SHOW A BAND THAT CANNOT BE JUSTIFIED

Coordination returns `risk: "unbanded"` when no matched organic baseline is configured. The interface must show the magnitude and print the caveat, and must **not** colour it as low/medium/high or imply severity. Absolute thresholds previously banded ordinary organic traffic as "high risk"; that is the failure this prevents.

RULE 2 – A GAP IS NOT A ZERO

Null and zero are different and must look different. No engagement data means reach cannot be split — not 0% manufactured. Unscored sentiment is not neutral sentiment. A thin corpus is not "no actors found".

RULE 3 – PARTIAL RESULTS STAY ON SCREEN

If the analysis fails after documents have been collected, those documents are real evidence. The interface shows them with an honest banner. It must not replace the screen with a generic error.

RULE 4 – VIEWS ARE IMPRESSIONS, NOT ACTIONS

Views are never summed into engagement. A single video's view count would otherwise outweigh every deliberate interaction in the corpus. Show both, labelled distinctly.

RULE 5 – SAY WHAT WAS NOT ESTABLISHED

Every screen with a conclusion should be able to state its limits: which sources failed, how many documents were undated, what the assessment does not cover. This is a feature of the product, not a footnote to hide.

RULE 6 – SYNDICATION IS NOT AMPLIFICATION

One wire story carried by six outlets is normal publishing. It is reported as context, never as a coordination finding, and the interface should reflect that difference in weight.

Every state to design

The interface spends much of its life in a state other than "everything worked". These are not edge cases.

STATE	WHEN	WHAT IT MUST CONVEY
Collecting	0-8s	Which sources have answered so far. Not a spinner.
Evidence ready, analysis running	8-35s	The documents are final and usable. The assessment is coming. This is a legitimate working state, not a loading screen.
Complete	~35s	
Degraded	Common	Some sources failed; the result is real but partial. Which ones, and what that means for the conclusion.
Thin corpus	Common	Fewer than 5-8 on-topic documents. Scores are suppressed by design. Must read as "too little evidence", not "nothing found".
Nothing on topic	Occasional	Documents were collected but none matched. Show what was collected and dropped, so the user can judge whether the gate was wrong.
Analysis failed	Occasional	Evidence stays; assessment is absent and says so.
Restored case	Any time	Viewing a capture from the past. Age must be unmistakable — a reading from three hours ago must never be mistaken for a live one.
Partial case	Any time	A saved case whose analysis never completed. Evidence only.
Unbanded	Currently always	Coordination magnitude with no risk level. See Rule 1.

Technical constraints

- **The frontend is one file** — `templates/index.html`, ~5,400 lines of vanilla JS, HTML and CSS with no build step and no framework. A rebuild could change that, but it is a real decision with deployment consequences, not a detail.
- **Two-phase delivery** — the main screen consumes a server-sent event stream, not a single JSON response. Any design must work with content arriving in two waves.
- **Case storage is browser-local** — IndexedDB for payloads, localStorage for the index. Cases do not follow the user to another device.
- **No external requests from the page** beyond the app's own origin and Google Fonts.
- **Dark and light both required.** The current interface is dark-only.
- **Right-to-left content is routine** — Arabic, Persian and Hebrew documents appear in normal use and need proper bidirectional handling, not a Latin layout with foreign text dropped in.
- **Print matters** — analysts export. There is already a print-first report route.

Open questions for the designer

Genuinely undecided. Answers will shape the build more than any styling choice.

1. **What is the primary unit of the main screen** — the assessment, the distinct claim, or the actor? Each produces a different product.
2. **How much interpretation belongs on first load?** The first delivery arrives 25 seconds before the analysis. Does the screen open on evidence and grow a conclusion, or hold a space for it?
3. **Where do limitations live?** Inline with each figure, in a persistent panel, or gated behind disclosure? They must be visible before the user repeats the claim.
4. **Is this one screen or a workspace?** Analysts run many queries per session and compare them. Nothing today supports comparison.
5. **How does time enter the product?** Narrative tracking gives stable identity across observations. That is the platform's strongest unused asset and has no home in the interface.

Written from the deployed codebase, not from a description of it. Figures — 96 kept of 199 collected, 9 of 21 sources, ~8s to first screen — are from a live run of `#covid1948` on 4 September 2026. Anything marked `GAP` is supported by data the backend already produces.